

E.G.S. PILLAY ENGINEERING COLLEGE, (Autonomous)

NAGAPATTINAM – 611 002.

(Affiliated to Anna University, Chennai | Accredited by NAAC with 'A++' Grade
Accredited by NBA | Approved by AICTE, New Delhi)



REGULATIONS - R2023

B.E. / B.Tech. – SECOND SEMESTER CURRICULUM

Second Semester										
S.No	COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX. MARKS		
								CA	ES	TOTAL
Theory Courses										
1	2301MA203	Statistics and Numerical Methods	BSC	3	1	0	4	40	60	100
2	2301PH202	Materials Science	BSC	3	0	0	3	40	60	100
3	2301GEX01	Foundation of Electrical and Electronics Engineering	ESC	3	0	0	3	40	60	100
4	2301GEX02	Engineering Graphics	ESC	2	1	0	3	40	60	100
5		Language Elective	EEC	2	0	0	2	100	0	100
6	2301TA201	Tamils and Technology / தமிழரும் தொழில்நுட்பமும்	HSMC	1	0	0	1	100	0	100
Laboratory Courses										
7	2301HS251	Engineering Exploration	HSMC	0	0	2	1	100	0	100
8	2301GE251	CAD Laboratory	ESC	0	0	2	1	100	0	100
9	2301GEX51	Computer Practices Laboratory	ESC	0	0	2	1	100	0	100
10	2301GEX53	Foundation of Electrical and Electronics Engineering Laboratory	ESC	0	0	2	1	60	40	100
11	2301LS201	Life Skills course – II		0	0	0	0	100	0	100
			TOTAL	14	2	8	20	820	280	1100

MODULEIII	SOLUTION OF EQUATIONS AND EIGENVALUE PROBLEMS	9Hours
Solution of algebraic and transcendental equations - Fixed point iteration method – Newton Raphson method- Solution of linear system of equations - Gauss elimination method – Pivoting - Gauss Jordan method – Iterative methods of Gauss Jacobi and Gauss Seidel - Eigen values of a matrix by Power method and Jacobi’s method for symmetric matrices.		
MODULEIV	INTERPOLATION, NUMERICAL DIFFERENTIATION AND NUMERICAL INTEGRATION	9Hours
Lagrange’s and Newton’s divided difference interpolations – Newton’s forward and backward difference interpolation – Approximation of derivatives using interpolation polynomials – Numerical single and double integrations using Trapezoidal and Simpson’s 1/3 rules.		
MODULEV	NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS	9Hours
. Single step methods: Taylor’s series method - Euler’s method - Modified Euler’s method - Fourth order Runge - Kutta method for solving first order differential equations - Multi step methods: Milne’s and Adams - Bash forth predictor corrector methods for solving first order differential equations		
TOTAL:45+15=60HOURS		
REFERENCES:		
1. G.B.Thomas and R.L. Finney, Calculus and Analytic geometry,9 th Edition, Pearson, Reprint, 2002.		
2. Erwin Kreyszig, Advanced Engineering Mathematics,9 th Edition,John Wiley& Sons,2006.		
3. W.E.Boyce and R.C.DiPrima, Elementary Differential Equations and Boundary Value Problems, 9th Edn, Wiley India, 2009.		
4. S.L.Ross,Differential Equations,3 rd Ed.,Wiley India,1984		
5. J.W.Brown and R.V.Churchill, Complex Variables and Applications,7 th Ed., McGrawHill, 2004.		
6. N.P.Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.		
7. B.S.Grewal,Higher Engineering Mathematics, Khanna Publishers,36 th Edition,2010.		

2301PH202	MATERIALS SCIENCE	L	T	P	C						
	(Common to Mechanical & Civil)	3	0	0	3						
PREREQUISITE:											
	1. Basic knowledge in Solid State Physics										
COURSE OBJECTIVES:											
CO 1:	To make the students to understand the basics of crystallography and its importance in studying materials properties.										
CO 2:	To understand the electrical properties of materials including free electron theory, applications of quantum mechanics and magnetic materials.										
CO 3:	To instil knowledge on physics of semiconductors, determination of charge carriers and device applications										
CO 4:	To establish a sound grasp of knowledge on different optical properties of materials, optical displays and applications										
CO 5:	To inculcate an idea of significance of nano structures, quantum confinement and ensuing nano device applications.										
COURSE OUTCOMES:											
On the successful completion of the course, students will be able to											
CO1:	Understand the basics of crystallography and its importance for varied materials properties										
CO2:	Differentiate between the electrical and magnetic properties of materials and their applications										
CO3:	Apply the concepts of semiconductor physics in functioning of semiconductor devices										
CO4:	Apply the properties of dielectric materials and working principles of various devices										
CO5:	Understand the importance of functional nanoelectronic devices.										
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	1	1	1					
CO2	3	3	2	2	2	1					1
CO3	3	3	1	1	2	1					
CO4	3	3	2	2	2	1					
CO5	3		2	2	2	1					
COs Vs PSOs MAPPING											
COURSE CONTENTS:											
MODULE I	CRYSTALLOGRAPHY									9 Hours	
Crystal structures: BCC, FCC and HCP – directions and planes - linear and planar densities – crystal imperfections- edge and screw dislocations – grain and twin boundaries - Burgers vector and elastic strain energy- Slip systems, plastic deformation of materials - Polymorphism – phase changes – nucleation and											

growth – homogeneous and heterogeneous nucleation.		
MODULE II	ELECTRICAL AND MAGNETIC PROPERTIES OF MATERIALS	9 Hours
Classical free electron theory - Expression for electrical conductivity – Thermal conductivity, expression - Quantum free electron theory : Tunneling – degenerate states – Fermi- Dirac statistics – Density of energy states – Electron in periodic potential – Energy bands in solids - Magnetic materials: Dia, para and ferromagnetic effects – paramagnetism in the conduction electrons in metals – exchange interaction and ferromagnetism – quantum interference devices – GMR devices.		
MODULE III	SEMICONDUCTORS AND TRANSPORT PHYSICS	9 Hours
Intrinsic Semiconductors – Energy band diagram – direct and indirect band gap semiconductors – Carrier concentration in intrinsic semiconductors – extrinsic semiconductors - Carrier concentration in N-type & P-type semiconductors – Variation of carrier concentration with temperature – Carrier transport in Semiconductors: Drift, mobility and diffusion – Hall effect and devices – Ohmic contacts – Schottky diode.		
MODULE IV	DIELECTRIC PROPERTIES OF MATERIALS	9 Hours
Polarization mechanisms: electronic, ionic, orientational, interfacial and total polarization – frequency dependence – local field/ Internal field derivation and Clausius-Mossetti equation – dielectric constant and dielectric loss.		
MODULE V	NANOELECTRONIC DEVICES	9 Hours
Quantum confinement – Quantum structures – quantum wells, wires and dots – Zener-Bloch oscillations – Resonant tunneling – quantum interference effects - mesoscopic structures - Single electron phenomena – Single electron Transistor. Semiconductor photonic structures – 1D, 2D and 3D photonic crystal. Active and passive optoelectronic devices – photo processes – spintronics – carbon nanotubes: Properties and applications.		
TOTAL: 45 HOURS		
REFERENCES:		
1. V.Raghavan. Materials Science and Engineering: A First Course, Prentice Hall India Learning Private Limited, 2015.		
2. S.O. Kasap, Principles of Electronic Materials and Devices, Mc-Graw Hill, 2018		
3. Jasprit Singh, Semiconductor Devices: Basic Principles, Wiley (India), 2007.		
4. Jasprit Singh, Semiconductor Optoelectronics: Physics and Technology, Mc-Graw Hill India (2019)		
5. G.W.Hanson. Fundamentals of Nanoelectronics. Pearson Education (Indian Edition), 2009.		
6. https://archive.nptel.ac.in/courses/108/108/108108122/		
7. https://onlinecourses.nptel.ac.in/noc20_ph24/preview		

2301GEX01	FOUNDATION OF ELECTRICAL AND ELECTRONICS ENGINEERING							L	T	P	C
(Common to CSE, IT, AIDS, BME, MECH, CIVIL)								3	0	0	3
PREREQUISITE:											
1. Physics											
COURSE OBJECTIVES:											
CO 1:		To introduce basic DC and AC circuits									
CO 2:		To impart knowledge in the basic working principles and applications of electrical machines and measuring instruments									
CO 3:		To educate the fundamental concepts of analog and digital electronics.									
COURSE OUTCOMES:											
On the successful completion of the course, students will be able to											
CO1:		Acquire basic knowledge on DC, AC circuits and wiring.									
CO2:		Understand the construction, working principle and applications of Electrical Machines.									
CO3:		Understand the various measuring instruments and concepts of transducers.									
CO4:		Obtain the knowledge of semiconductor devices and their applications.									
CO5:		Acquire basic knowledge on logic gates and Boolean algebra.									
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3		3			1	1	3		2
CO2	3										1
CO3	3			1					3		1
CO4	3			1	3				3		1
CO5	3	1		2	3				3		1
COs Vs PSOs MAPPING											
				COs	PSO1	PSO2	PSO3				
				CO1	3		3				
				CO2	3						
				CO3	3						
				CO4		3	3				
				CO5		3	3				
COURSE CONTENTS:											
MODULE I		ELEMENTARY CIRCUIT CONCEPTS								9 Hours	
Introduction to DC and AC circuits - Ohm's Law, Kirchhoff's Laws, Simple problems; Mesh analysis, Nodal Analysis; Generation of AC waveform - average value, RMS value, form factor, peak factor; Introduction to three phase systems; Electrical safety (not for examination)											
MODULE II		ELECTRICAL MACHINES								9 Hours	
Construction, working principle, EMF equation, types and applications of DC Generators, working principle of DC Motors, Torque equation, Types and application. Working principle and applications of single phase transformers and single phase induction motors, three phase alternator.(Simple approach)											
MODULE III		MEASURING INSTRUMENTS								9 Hours	
Measuring instruments; Classification of instruments -PMMC, MI instruments, dynamometer type wattmeter, static watt-hour meter; CRO- Principle and operation; Introduction to transducers- RTD, LVDT.											

MODULE IV	ANALOG ELECTRONICS	9 Hours
Semiconductor devices- V-I characteristics of PN junction diode and Zener diode; Rectifiers - Half wave and full wave rectifiers; BJT, SCR, MOSFET construction and operation (simple approach)		
MODULE V	DIGITAL ELECTRONICS	9 Hours
Binary Number System; Logic Gates; Boolean algebra; De-Morgan's theorem; Half and Full Adder. SOP and POS forms, K-map representations - minimization using K maps (Simple Problems only)		
TOTAL: 45 HOURS		
REFERENCES:		
1. Mittle N., "Basic Electrical Engineering", Tata McGraw Hill Edition, New Delhi, 1990.		
2. Sedha R.S., "Applied Electronics", S. Chand & Co., 2006.		
3. Smarajit Ghosh, "Fundamentals of Electrical and Electronics Engineering", 2 nd Edition, PHI Learning, 2010.		
4. R. Muthusubramaniam, S. Salaivahanan and K.A. Mureleedharan, "Basic Electrical Electronics and Computer Engineering", Tata McGraw Hill, 2004		
5. D.P. Kothari and I.J. Nagrath, "Theory and Problems of Basic Electrical Engineering", PHI learning, New Delhi, 2004.		
6. J.B. Gupta, "Fundamentals of Electrical Engineering and Electronics", S.K. Kataria and Sons, Reprint 2012 Edition		
7. R.L. Boylestad and L. Nashelsky, "Electronic Devices and Circuit Theory", Pearson, 11th Edition, 2013.		
8. Donald P. Leach, Albert Paul Malvino and Goutam Saha, "Digital Principles and Applications", McGraw-Hill Education, 8th Edition, 2014.		

2301GEX02	ENGINEERING GRAPHICS										L	T	P	C
											2	1	0	3
Prerequisite:														
1. Basic knowledge about geometry														
2. Lettering and Dimensioning														
COURSE OBJECTIVES:														
CO 1:	To develop in students, graphic skills for communication of concepts, ideas and design of Engineering products													
CO 2:	To expose them to existing national standards related to technical drawings													
COURSE OUTCOMES:														
On the successful completion of the course, students will be able to														
CO1:	Construct conic curves, involutes and cycloids													
CO2:	Solve problems involving projection of points, lines and plane surfaces													
CO3:	Draw the projection of a sectioned simple solids													
CO4:	Draw the development of a sectioned simple solids													
CO5:	Draw the orthographic, isometric projection of simple solids													
COs Vs POs / PSOs MAPPING:														
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	1	2		2					3	2	2	2	
CO2	3	1	2		2					3	2	2	2	
CO3	3	1	2		2					3	2	2	2	
CO4	3	1	2		2					3	2	2	2	
CO5	3	1	2		2					3	2	2	2	
COURSE CONTENTS:														
MODULE I	BASIC CONCEPTS OF TECHNICAL DRAWING AND PLANE CURVES												9 Hours	
Importance of graphics in engineering applications – Use of drafting instruments – BIS conventions and specifications – Size, Scale, layout and folding of drawing sheets – Lettering and dimensioning. Basic Geometrical constructions, Curves used in engineering practices: Conics – Construction of ellipse, parabola and hyperbola by eccentricity method – Construction of cycloid – construction of involutes of square and circle – Drawing of tangents and normal to the above curves.														
MODULE II	PROJECTION OF POINTS, LINES AND PLANE SURFACES												9 Hours	
Principal Planes-First angle projection-projection of points. Projection of straight lines (only First angle projections) inclined to both the principal planes - Determination of true lengths and true inclinations by rotating line method and traces. Projection of planes (polygonal and circular surfaces) inclined to both the principal planes by rotating object method.														
MODULE III	PROJECTION OF SOLIDS												9 Hours	
Projection of simple solids like prisms, pyramids, cylinder and cone when the axis is inclined to one of the principal planes by rotating object method.														
MODULE IV	PROJECTION OF SECTIONED SOLIDS AND DEVELOPMENT OF SURFACES												9 Hours	
Sectioning of above solids in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other – obtaining true shape of section. Development of lateral surfaces of simple and sectioned solids – Prisms, pyramids cylinders and cones.														

MODULE V	ORTHOGRAPHIC AND ISOMETRIC PROJECTION	9 Hours
Visualization concepts–Representation of Three-Dimensional objects – Layout of views- Free hand sketching of multiple views from pictorial views of Objects. Isometric view - Prisms, pyramids, cylinders, cones. Principles of isometric projection – isometric scale – Isometric projections of simple solids and truncated solids - Prisms, pyramids, cylinders, cones- combination of two solid objects in simple vertical positions and miscellaneous problems.		
TOTAL: 45 HOURS		
REFERENCES:		
1. Bhatt N.D. and Panchal V.M., Charotar Publishing House, 53rd Edition, 2019.		
2. Natrajan K.V., A Text Book of Engineering Graphics, Dhanalakshmi Publishers, Chennai, 2018.		
3. Parthasarathy, N. S. and Vela Murali, “Engineering Drawing”, Oxford University Press, 2015.		
4. Basant Agarwal and Agarwal C.M., “Engineering Drawing”, McGraw Hill, 2nd Edition, 2019.		
5. Gopalakrishna K.R., “Engineering Drawing” (Vol. I&II combined), Subhas Publications, Bangalore, 27th Edition, 2017		
6. Shah M.B., and Rana B.C., “Engineering Drawing”, Pearson Education India, 2 nd Edition, 2009.		
7. Venugopal K. and Prabhu Raja V., “Engineering Graphics”, New Age International (P) Limited, 2008.		

2301TA201	தமிழரும் தொழில்நுட்பமும்/ TAMIL AND TECHNOLOGY										L	T	P	C	
											1	0	0	1	
PRE-REQUISITE:															
	The Tamils living in different parts of the World need to keep in touch with the motherland and the mother tongue and be knowledgeable about their heritage in order to preserve their cultural identity and observe their traditional and cultural activities.														
	Recognizing this fact and for meeting the felt and emerging needs of the Tamil Communities and others interested in Tamil studies														
COURSE OBJECTIVES:															
	Tamil Literature is way of a life. It focuses on the historical significance of ethics, moral culture in the Tamil context.														
	Tamil Modern literature emphasizes on the modern development of the behavioral, moral and ethical														
	Technology is the important key for a language and a new sector for the students to voice out for a social cause														
COURSE OUTCOMES:															
At the end of this course, Students will be able to,															
CO1:	Develop a spirit of patriotism.														
CO2:	Understand the plight of the people living in the society and Biological Struggles.														
CO3:	Remember the life style of the Sangam people and To recognize the heroic spirit of the ancient Tamil kings														
CO4:	Evaluate the quality and morals of local life through Tamil literature														
CO5:	Introducing the various Literary Genres and dramas and enable them to produce innovative ideas in modern literary theories														
COs Vs POs / PSOs MAPPING:															
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
	CO1									3					
	CO2									3					
	CO3									3					
	CO4									3					
	CO5									3					
COURSE CONTENTS:															
MODULE I	WEAVING AND CERAMIC TECHNOLOGY												3 Hours		
Weaving Industry during Sangam Age–Ceramic technology–Black and Red Ware Potteries (BRW) Graffition Potteries.															
அலகு I நெசவு மற்றும் பானைத் தொழில்நுட்பம்: 3															
சங்க காலத்தில் நெசவுத் தொழில் – பானைத் தொழில்நுட்பம் - கருப்பு சிவப்பு பாண்டங்கள் – பாண்டங்களில் கீறல் குறியீடுகள்.															
MODULE II	DESIGN AND CONSTRUCTION TECHNOLOGY												3 Hours		
Designing and Structural construction House & Designs in house hold materials during Sangam Age Building materials and Hero stones of Sangam age -Details of Stage Constructions in Silappathikaram - Sculptures and Temples of Mamallapuram - Great Temples of Cholas and other worship places - Temples of Nayaka Period - Type study (Madurai Meenakshi Temple) - Thirumalai Nayakar Mahal – Chetti Nadu Houses, Indo-Saracenic architecture at Madras during British Period.															

அலகு II	வடிவமைப்பு மற்றும் கட்டிடத் தொழில்நுட்பம்: சங்க காலத்தில் வடிவமைப்பு மற்றும் கட்டுமானங்கள் & சங்க காலத்தில் வீட்டுப் பொருட்களில் வடிவமைப்பு- சங்க காலத்தில் கட்டுமான பொருட்களும் நடுகல்லும் – சிலப்பதிகாரத்தில் மேடை அமைப்பு பற்றிய விவரங்கள் - மாமல்லபுரச் சிற்பங்களும், கோவில்களும் – சோழர் காலத்துப் பெருங்கோயில்கள் மற்றும் பிற வழிபாட்டுத் தலங்கள் – நாயக்கர் காலக் கோயில்கள் - மாதிரி கட்டமைப்புகள் பற்றி அறிதல், மதுரை மீனாட்சி அம்மன் ஆலயம் மற்றும் திருமலை நாயக்கர் மஹால் – செட்டிநாட்டு வீடுகள் – பிரிட்டிஷ் காலத்தில் சென்னையில் இந்தோ-சாரோசெனிக் கட்டிடக் கலை.	3
MODULE III	MANUFACTURING TECHNOLOGY	3 Hours
	Art of Ship Building - Metallurgical studies - Iron industry - Iron smelting, steel - Copper and gold - Coins as source of history - Minting of Coins – Beads making – industries Stone beads - Glass beads – Terra-cotta beads – Shell beads/bone beads – Archeological evidences –Gems tone types described in Silappathikaram. அலகு III உற்பத்தித் தொழில் நுட்பம்: கப்பல் கட்டும் கலை – உலோகவியல் – இரும்புத் தொழிற்சாலை – இரும்பை உருக்குதல், எஃகு – வரலாற்றுச் சான்றுகளாக செம்பு மற்றும் தங்க நாணயங்கள் – நாணயங்கள் அச்சடித்தல் – மணி உருவாக்கும் தொழிற்சாலைகள் – கல்மணிகள், கண்ணாடி மணிகள் – சுடுமண் மணிகள் – சங்கு மணிகள் – எலும்புத்துண்டுகள் – தொல்லியல் சான்றுகள் – சிலப்பதிகாரத்தில் மணிகளின் வகைகள்.	3
MODULE IV	AGRICULTURE AND IRRIGATION TECHNOLOGY	3 Hours
	Dam, Tank, ponds, Sluice, Significance of Kumizhi Thoompu of Chola Period, Animal Husbandry -Wells designed for cattle use - Agriculture and Agro Processing - Knowledge of Sea - Fisheries –Pearl – Conche diving – Ancient Knowledge of Ocean – Knowledge Specific Society அலகு IV வேளாண்மை மற்றும் நீர்ப்பாசனத் தொழில் நுட்பம்: அணை, ஏரி, குளங்கள், மதகு – சோழர்காலக் குழுமித் தூம்பின் முக்கியத்துவம் – கால்நடை பராமரிப்பு – கால்நடைகளுக்காக வடிவமைக்கப்பட்ட கிணறுகள் – வேளாண்மை மற்றும் வேளாண்மைச் சார்ந்த செயல்பாடுகள் – கடல்சார் அறிவு – மீன்வளம் – முத்து மற்றும் முத்துக்குளித்தல் – பெருங்கடல் குறித்த பண்டைய அறிவு – அறிவுசார் சமூகம்.	3
MODULE V	SCIENTIFIC TAMIL & TAMIL COMPUTING	3 Hours
	Development of Scientific Tamil – Tamil computing – Digitalization of Tamil Books – Development of Tamil Software – Tamil Virtual Academy – Tamil Digital Library – Online Tamil Dictionaries –Sorkuvai Project. அலகு V அறிவியல் தமிழ் மற்றும் கணித்தமிழ்: அறிவியல் தமிழின் வளர்ச்சி –கணித்தமிழ் வளர்ச்சி - தமிழ் நூல்களை மின்பதிப்பு செய்தல் – தமிழ் மென்பொருட்கள் உருவாக்கம் – தமிழ் இணையக் கல்விக்கழகம் – தமிழ் மின் நூலகம் – இணையத்தில் தமிழ் அகராதிகள் – சொற்குவைத் திட்டம்.	3
TOTAL:15 HOURS		
REFERENCES:		

1. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
2. Social Life of the Tamils - The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.
3. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu) (Published by: International Institute of Tamil Studies).
4. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies.)
5. Keeladi - ‘Sangam City Civilization on the banks of river Vaigai’ (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
6. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by: The Author)
7. Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
8. Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) – Reference Book.

2301HS251	ENGINEERING EXPLORATION										L	T	P	C
											0	0	2	1
Prerequisite:														
1. Basic knowledge about Engineering products														
2. Willingness to design and fabricate simple products														
COURSE OBJECTIVES:														
CO 1:	Build mindsets & foundations essential for designers													
CO 2:	Learn about the Human-Centered Design methodology and understand their real-world applications													
CO 3:	Use Design Thinking for problem solving methodology for investigating ill-defined problems.													
CO 4:	Undergo several design challenges and work towards the final design challenge													
CO 5:	Apply Design Thinking on the following Streams to Mechanical tools, Eco-friendly solutions for waste management, infrastructure, safety, alternative energy sources, Agriculture, Environmental science, Robotics and other fields of Engineering.													
COURSE OUTCOMES:														
On the successful completion of the course, students will be able to														
CO1:	Describe class objectives & harness the designer mindset													
CO2:	Describe the design thinking methodology													
CO3:	Demonstrate about teams & problems through this													
CO4:	Demonstrate about empathy													
CO5:	Demonstrate about how to brainstorm effectively													
CO6:	Design and fabricate effective prototypes													
CO7:	Inspect prototypes and ideas through user feedback													
CO8:	Prepare final Report and Presentation													
COs Vs POs / PSOs MAPPING:														
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	1				2			3	3	2			
CO2	2	1				2			3	3	2			
CO3	2	1				2			3	3	2			
CO4	2	1				2			3	3	2			
CO5	2	1				2			3	3	2			
CO6	3	3	2		2	2			3	3	2			
CO7	3	3	3		2	2			3	3	2			
CO8	2	1			1	2			3	3	2			
COURSE CONTENTS:														
HOW TO PURSUE THE PROJECT WORK?														
1. The first part will be learning-based-masking students to embrace the methodology by exploring all the phases of design thinking through the wallet/ bag challenge and podcasts.														
2. The second part will be more discussion-based and will focus on building some necessary skills as designers and learning about complementary material for human- centered design.														
3. The class will then divide into teams and they will be working with one another for about 2 – 3 weeks. These teams and design challenges will be the basis for the final project and final presentation to be presented.														
4. The teams start with Design Challenge and go through all the phases more in depth from coming up														

with the right question to empathizing to ideating to prototyping and to testing.

5. Outside of class, students will also be gathering the requirements, identifying the challenges, usability, importance etc
6. At the end, Students are required to submit the final reports, and will be evaluated by the faculty.

TASKS TO BE DONE:

Task 1: Everyone is a Designer

- Understand class objectives & harness the designer mindset

Task 2: The Wallet/Bag Challenge and Podcast

- Gain a quick introduction to the design thinking methodology
- Go through all stages of the methodology through a simple design challenge
- Podcast: Observe, Listen and Engage with the surrounding environment and identify a design challenge.

Task 3: Teams & Problems

- Start Design Challenge and learn about teams & problems through this
- Foster team collaboration, find inspiration from the environment and learn how to identify problems

Task 4: Empathizing

- Continue Design Challenge and learn empathy
- Learn techniques on how to empathize with users
- Go to the field and interview people in their environments
- Submit Activity Card

Task 5: Ideating

- Continue Design Challenge and learn how to brainstorm effectively
- Encourage exploration and foster spaces for brainstorming
- Submit Activity Card

Task 6: Prototyping

- Continue Design Challenge and learn how to create effective prototypes
- Build tangible models and use them as communication tools
- Start giving constructive feedback to classmates and teammates
- Submit Activity Card

Task 7: Testing

- Finish Design Challenge and iterate prototypes and ideas through user feedback
- Evolve ideas and prototypes through user feedback and constructive criticism
- Get peer feedback on individual and group performance
- Submit Activity Card

Task 8:

- Final Report Submission and Presentation
- Method of Evaluation: Same as Mini project category. Project exhibition may be conducted.

TOTAL: 45 HOURS

REFERENCES:

1. Tom Kelly, The Art of Innovation: Lessons in Creativity From IDEO, America's Leading Design Firm (Profile Books, 2002)
2. Tim Brown, Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation (HarperBusiness, 2009)
3. Jeanne Liedtka, Randy Salzman, and Daisy Azer, Design Thinking for the Greater Good: Innovation in the Social Sector (Columbia Business School Publishing, 2017)

OTHER USEFUL DESIGN THINKING FRAMEWORKS AND METHODOLOGIES:

1. Human-Centered Design Toolkit (IDEO); <https://www.ideo.com/post/design-kit>
2. Design Thinking Boot Camp Bootleg (Stanford D-School); <https://dschool.stanford.edu/resources/the-bootcamp-bootleg>
3. Collective Action Toolkit (frogdesign); https://www.frogdesign.com/wpcontent/uploads/2016/03/CAT_2.0_English.pdf
4. Design Thinking for Educators (IDEO); <https://designthinkingforeducators.com/>

2301GE251	CAD LABORATORY										L	T	P	C	
											0	0	2	1	
Prerequisite:															
1. Basic Computer knowledge															
2. Engineering Graphics															
COURSE OBJECTIVES:															
CO 1:	To develop in students, Drafting skills for communication of concepts, ideas and design of Engineering products														
CO 2:	To expose them to existing national standards related to technical drawings														
COURSE OUTCOMES:															
On the successful completion of the course, students will be able to															
CO1:	Ability to use the software packers for drafting and modeling														
CO2:	Learned basic concept to drawing, edit, dimension, hatching etc. to develop 2& 3D Modelling.														
CO3:	Able to create front view and top view of simple solids														
CO4:	Able to create isometric projection of simple objects.														
CO5:	Able to Create 3D models of Simple Objects and obtaining 2-D multi-view drawings from 3-D model														
COs Vs POs / PSOs MAPPING:															
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
	CO1	3	-	-	-	3	-	-	-	2	2	2	-	-	3
	CO2	3	-	-	-	3	-	-	-	2	2	2	-	-	3
	CO3	3	-	-	-	3	-	-	-	2	2	1	-	-	3
	CO4	3	-	-	-	3	-	-	-	2	2	1	-	-	3
	CO5	3	-	-	-	3	-	-	-	2	2	1	-	-	3
General:															
Study of basics commands of a CAD (AutoCAD) software - two-dimensional drawing, editing, layering and dimensioning - coordinate Systems – units – limits															
List of Experiments:															
1. Construction of Lines, Simple geometries, and Title Block with necessary text and projection symbol.															
2. Construction of Ellipse, Parabola, Hyperbola.															
3. Construction of cycloids and involutes.															
4. Construction of Projection of a straight Line.															
5. Draw the orthographic views (front, top, and side views) of simple solids.															
6. Draw sectional views of prism, pyramid, cylinder, cone, etc,															
7. Creation of 3-D models of simple objects.															
														TOTAL: 45 HOURS	
REFERENCES:															
1. N.D. Bhatt, Machine Drawing, Charotar Publishing House Pvt. Ltd., 2014.															
2. P.S. Gill, A Textbook of Machine Drawing, Katson books, 2013.															
3. R.K. Dhawan, A Textbook of Machine Drawing, S. Chand,2012.															
4. K.C. John, Textbook of Machine Drawing, PHI Learning Pvt. Ltd.,2009.															

2301GEX51	COMPUTER PRACTICES LABORATORY										L	T	P	C	
											0	0	2	1	
PREREQUISITE:															
There is no prerequisite for the course															
COURSE OBJECTIVES:															
1.	To be familiar with Computer Hardware Components and installation of software.														
2.	Make use of office package and to be familiar with the use of Office software.														
3.	To learn about searching, downloading, and storing contents in the Cloud Network.														
COURSE OUTCOMES:															
Upon the successful completion of the course, students will be able to															
CO1	Perform assembling and disassembling of desktop machine with different peripheral and software installation and servicing.														
CO2	Simulate data using MS office for Presentation and Visualization.														
CO3	Use browsers for searching & accessing/storing the contents to/from cloud.														
LIST OF EXPERIMENTS:															
1. Familiarization of Computers & Computer Hardware Components															
2. Familiarization of major types of storage/memory technology															
3. Installing various operating systems including software download/installation, Familiarization of basic software/tools															
4. Working with MS-Office: MS Word, MS Excel, MS Powerpoint															
5. Familiarization of Computer Shortcut keys															
6. Mini Project-1: Assemble your computer and install an Operating System															
7. Basics of Internet, Web browsers and Content Searching & accessing/storing the contents to/from cloud including DropBox															
8. Familiarization of various types of security threats including virus															
9. Computer Ethics; Open Source way															
10. Mini Project-2: Document preparation using MS Word, Data Processing using MS Excel and Presentation using MS PowerPoint															
TOTAL: 30 HOURS															
COs Vs POs & PSOs MAPPING:															
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
	CO1	3	3	2	2	-	-	-	2	2	-	1	1	1	1
	CO2	3	3	2	2	2	-	-	-	-	1	1	1	1	1
	CO3	3	3	2	1	-	-	-	2	-	-	1	1	1	1
HARDWARE/SOFTWARE REQUIREMENT															
1. Standalone Desktop Computers with Internet Connectivity															
2. Office Package															
3. Operating System Packages															
REFERENCES:															
1. Kevin Wilson, “Computer Hardware: The Illustrated Guide to Understanding Computer Hardware”, 2021															
2. Kumar Bittu, “Mastering MS Office”, 2020															
3. Ajay Mittal &Anitha Goel, “Computer Fundamentals and Programming in C”, 2017															
4. https://nptel.ac.in/courses/106103068															
5. https://docs.oracle.com/cd/E19121-01/sf.x2100m2/819-6592-13/Chap1.html															
6. https://www.linkedin.com/learning/topics/microsoft-office															

2301GEX53	FOUNDATION OF ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY											L	T	P	C	
	(Common to CSE, IT, AIDS, BME, MECH, CIVIL)											0	0	2	1	
COURSE OBJECTIVES:																
	To provide practical exposure to basic electrical and electronic circuits and devices.															
	To develop skills in circuit construction, testing, measurement, and analysis using standard laboratory instruments.															
	To enable students to evaluate the performance of electrical machines, power electronic systems, and renewable energy applications with an emphasis on energy efficiency and safety.															
COURSE OUTCOMES:																
Upon the successful completion of the course, students will be able to																
CO1:	Apply electrical circuit laws to analyze and verify DC circuits and electronic components.															
CO2:	Construct and test electrical wiring and regulated power supply circuits following safety standards.															
CO3:	Analyze the characteristics and applications of semiconductor devices and integrated circuits.															
CO4:	Design, simulate, and evaluate basic electrical, electronic, and renewable energy systems.															
CO5:	Perform experiments, interpret results, and assess the performance of electrical machines and energy systems.															
LIST OF EXPERIMENTS:																
1. Verification of Ohm’s law and Kirchhoff’s laws													3 Hours			
2. Residential house wiring using fuse, switch, indicator, lamp and energy meter													3 Hours			
3. V-I characteristics of PN junction diode / Zener diode													3 Hours			
4. IC 555 and IC 741 based experiments													3 Hours			
5. Energy conservation demonstration experiment using energy meter													3 Hours			
6. Waveform generation and calculation of RMS and average values													3 Hours			
7. Design of 6V regulated power supply													3 Hours			
8. Verification of Logic gates													3 Hours			
9. Speed control of DC shunt motor.													3 Hours			
10. I – V Characteristics of Solar PV cell (Simulation approach)													3 Hours			
TOTAL: 30 HOURS																
COs Vs POs / PSOs MAPPING:																
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	
	CO1	3	2	-	-	3	-	-	-	2	2	2	-	-	-	
	CO2	3	2	-	-	3	-	-	-	2	2	2	-	-	-	
	CO3	3	2	-	-	3	-	-	-	2	2	1	-	-	-	
	CO4	3	2	-	-	3	-	-	-	2	2	1	-	-	-	
	CO5	3	2	-	-	3	-	-	-	2	2	1	-	-	-	
REFERENCES																
1. Edward Hughes, “ Electrical Technology,” Pearson Education																
2. D.P. Kothari and Nagrath“ Basic Electronics”, MH Education 2013.																
3. Paul Scherz and Simon Monk “Practical Electronics for inventors” Mc Graw Hill Publications 2013.																
4. https://nptel.ac.in/courses/122106025/																
5. https://em-coep.vlabs.ac.in/exp/speed-control-dc-motor/simulation.html																
6. https://de-iitr.vlabs.ac.in/exp/truth-table-gates/simulation.html																
7. Dr.T.Suresh Padmanabhan, Dr.M.Vinothkumar and Dr.S.Sivamani, “Foundation of Electrical and Electronics Engineering Laboratory Manual”, June 2023.																

